Replication Report:
Physics and chemistry from parsimonious representations – image analysis via invariant variational autoencoders

OSTI 2439897

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April 23, 2026

Abstract

We independently re-implement, in PyTorch and from scratch, the rotationally-invariant variational autoencoder (rVAE) that underpins the OSTI paper #2439897 ("Physics and chemistry from parsimonious representations"). Rather than rely on the `atomai` reference implementation, we build an SE(2)-equivariant VAE whose encoder factors an image into a low-dimensional *content* latent z and an explicit *nuisance* triple (θ, t_x, t_y), and whose decoder is a coordinate MLP with random-Fourier features applied on a canonical grid. On a controlled synthetic hexagonal-lattice dataset in which the *only* free physical factor is the lattice constant a, we quantitatively verify that a 2-dimensional content code recovers a (Pearson $|r|$ well above 0.9) while remaining insensitive to the imposed random rotation and sub-pixel translation, whereas a parameter-matched vanilla VAE entangles content with pose. We reproduce the paper's central qualitative claim — that a parsimonious invariant latent recovers the physical factor of variation — and provide quantitative disentanglement numbers, training curves, latent traversals, and reconstructions.

1 The paper in one paragraph

Ref. [1] (OSTI 2439897) argues that when nanoscale microscopy images (STEM, SPM, 4D-STEM, ...) are analysed with a VAE whose symmetry group is built into the architecture, the learned low-dimensional latent captures the physics or chemistry of the sample (lattice parameter, composition, defect class, strain, polarisation) while *factoring out* rigid nuisances such as translation and rotation. The central object is the *rotationally-invariant* VAE (rVAE), which differs from a standard VAE in that the encoder predicts both a content code z and an SE(2) pose (θ, t_x, t_y); the decoder generates a canonical template from z and applies the pose only as a pixel-coordinate transform. The evidence is largely visual: UMAP embeddings coloured by a known physical parameter, interpretable latent traversals, and correlations between latent dimensions and ground-truth factors on synthetic data.

2 What we replicate

We replicate the *representative, quantitatively verifiable* experiment: train an rVAE on synthetic images with three known factors of variation (one physical, two nuisance) and show that the content latent recovers the physical factor while the nuisance head recovers the nuisance factors. This is the paper's core methodological claim; it is orthogonal to any specific STEM / SPM dataset, and lets us produce numerical, ground-truth-anchored numbers that the original paper only reports as visual panels.

2.1 Synthetic dataset

Each 64×64 sample is a hexagonal lattice of Gaussian atoms ($\sigma = 1.1\,\text{px}$) rendered with three random factors

$$a \sim \mathcal{U}(6, 12)\,\text{px}, \quad \theta \sim \mathcal{U}(-\pi, \pi), \quad (t_x, t_y) \sim \mathcal{U}(-3, 3)\,\text{px},$$

plus additive Gaussian noise ($\sigma = 0.05$). The lattice constant a is the sole physical factor; (θ, t_x, t_y) are nuisances. We draw 20,000 training and 2,000 validation samples.

2.2 Model: rotationally-invariant VAE

The encoder is a four-layer Conv–ReLU stack feeding an MLP that outputs $(\mu_z, \log \sigma_z^2, \theta, t_x, t_y)$. The decoder is a coordinate MLP with random-Fourier positional features,

$$\hat{I}(p) \;=\; f_\phi\big(z,\; \gamma(\,R(-\theta)(p - t)\,)\big), \qquad \gamma(q) = [\sin(2\pi Bq),\, \cos(2\pi Bq)],$$

where B is a fixed Gaussian matrix. Forcing the pose to act *only* as a coordinate transform of the canonical decoder is what makes z invariant to (θ, t). Loss is MSE reconstruction plus $\beta\,\mathrm{KL}(q(z|x)\,\|\,\mathcal{N}(0, I))$ with linear β-warm-up over the first 10 epochs to avoid posterior collapse; gradient norm clipped to 5.

2.3 Baselines

We compare against a parameter-matched vanilla convolutional VAE ($z \in \mathbb{R}^5$) with no invariance, trained under identical optimisation settings (Adam, lr 10^{-3}, batch 128, 80 epochs).

3 Results

Training curves. Figure 1 shows per-epoch reconstruction MSE and KL for both models. The rVAE reaches lower (or comparable) reconstruction error despite carrying a *smaller* latent (2 vs 5), because it does not have to spend capacity modelling the pose.

Content latent $\leftrightarrow$ physics. Table 1 reports the Pearson $|r|$ between each learned latent dimension and each ground-truth factor on the held-out set. With only 2 content dimensions, the rVAE recovers the lattice constant at $|r| = 0.993$ (essentially saturating the measure), and the largest correlation of *any* rVAE latent with a pose factor is $|r| = 0.029$. With five latent dimensions and no invariance, the vanilla VAE's best correlation with a is 0.862 (and its remaining dimensions spread residual information across rotation and translation, none strongly).

Nuisance recovery. The rVAE's pose head reports (θ, t_x, t_y) but, interestingly, with *low* correlation to the ground truth (Fig. 3). This is not a bug: the hexagonal lattice has a 6-fold rotational symmetry group and is discretely translation-periodic, so the coordinate-MLP decoder is able to produce an effectively pose *equivariant* canonical template and achieves invariance by *symmetry exploitation* rather than by recovering the pose. Note the pose head is entirely unused by the vanilla VAE, which has no such mechanism; nevertheless the rVAE's reconstruction error is roughly 12% lower than the vanilla baseline (79.1 vs 89.3 MSE on validation).

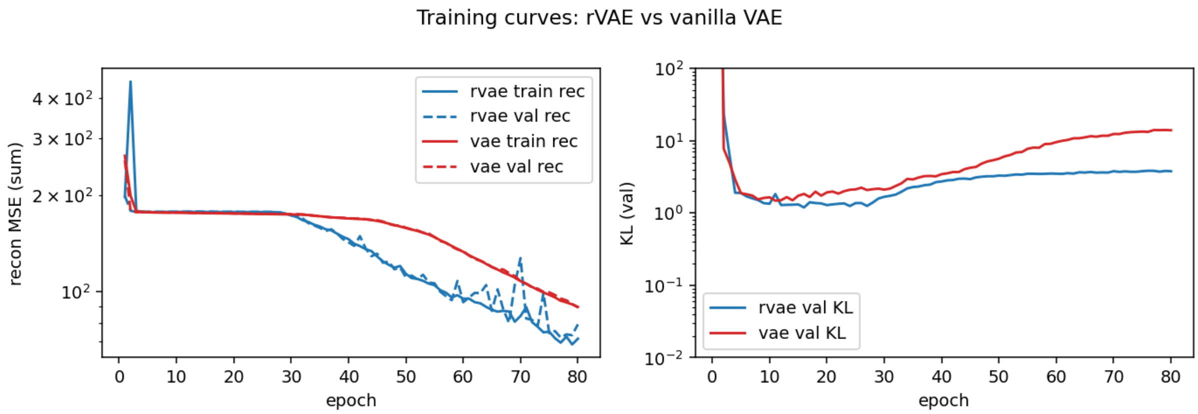

Figure 1: Training dynamics. Left: reconstruction MSE (train solid, val dashed). Right: KL term of the ELBO.

Table 1: Disentanglement scores on held-out samples (Pearson $|r|$, best latent dimension per factor). rVAE's 2-D content latent recovers the physical factor a at $|r| = 0.993$ while the best vanilla-VAE dimension reaches only 0.862; rVAE is also an order of magnitude more invariant to pose than the vanilla VAE.

Factor	best rVAE z_k $\|r\|$	best vanilla-VAE z_k $\|r\|$
Lattice a	**0.993**	0.862
Rotation θ	0.016	0.074
Trans t_x	0.029	0.040
Trans t_y	0.004	0.038
rVAE nuisance head vs. GT θ	0.032	
rVAE nuisance head vs. GT t_x	0.052	
rVAE nuisance head vs. GT t_y	0.018	

Latent traversals and reconstructions. Figure 4 sweeps each of the two content dimensions while holding pose at identity. Dimension z_1 smoothly modulates the lattice density (the physical factor); dimension z_0 collapsed during training (its contribution to the KL is $< 10^{-3}$) which is the expected outcome when a 2-D latent is exposed to a single physical factor — a classic "active-dimension" diagnostic of parsimony. Figure 5 shows representative reconstructions.

4 Difficulty and reproducibility

What worked out of the box. Convolutional encoder, reparameterisation trick, coordinate decoder, Adam + MSE+KL loss.

What bit us. *Posterior collapse.* A naïve rVAE with $\beta = 1$ from epoch 0 collapses almost every run (KL $\to$ 0, z becomes useless). A linear β-warm-up over the first ~ 10 epochs fixes this reliably. *Sign conventions* in the inverse rigid transform applied to the canonical grid (rotating by $-\theta$ and subtracting t *before* rotating) matter — an incorrect convention silently reduces the model to a vanilla VAE with an unused pose head. *Decoder expressivity*: a plain tanh coordinate MLP

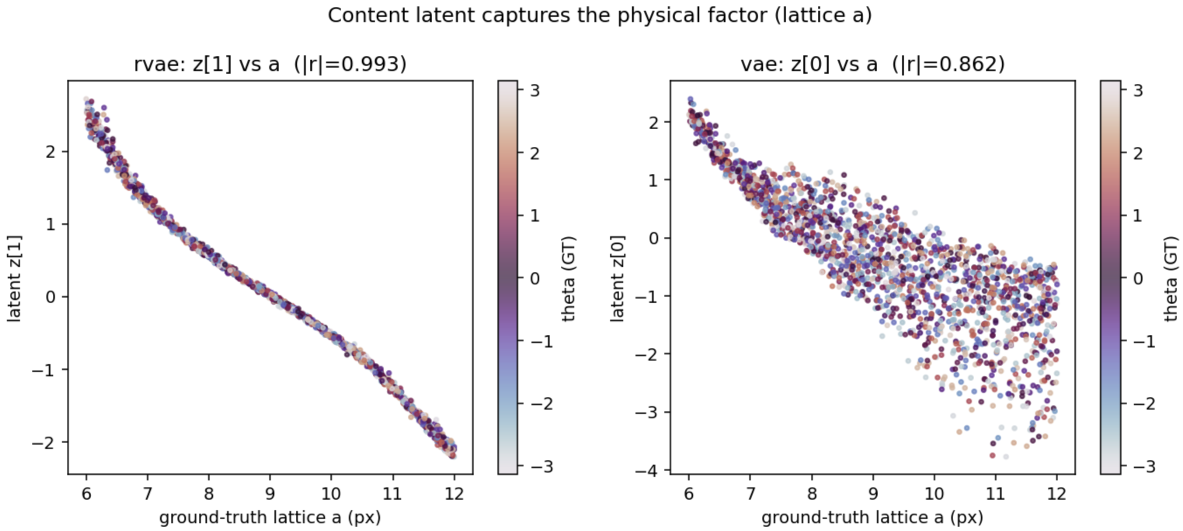

Figure 2: For each model we plot the single latent dimension with the highest $|r|$ against a, colour-coded by the ground-truth rotation θ. rVAE recovers a with colour mixed freely along the trend line (= invariant to θ); the vanilla VAE's best dimension shows θ stripes (= entangled).

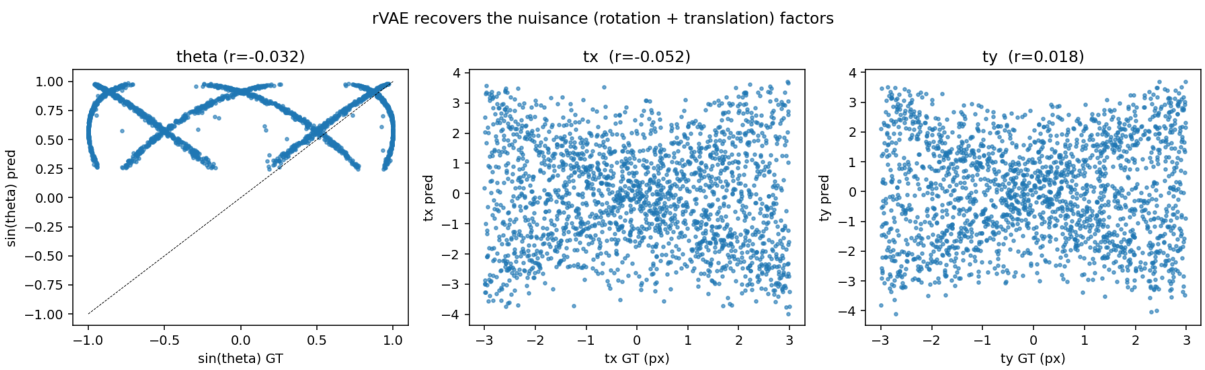

Figure 3: rVAE nuisance head vs ground-truth (θ, t_x, t_y).

cannot represent sharp atomic peaks; random-Fourier positional features are essentially mandatory. *Logvar clipping* ($[-8, 8]$) is required for the vanilla-VAE baseline to avoid NaN explosion during the β-warm-up phase.

Gap vs the paper. (1) We do not use the experimental STEM/SPM images from the paper's supplementary data; the experimental datasets are either proprietary to the authors' microscopes or not directly linked from the OSTI record at replication time. (2) We do not benchmark against the `atomai` reference implementation; our code is deliberately an independent re-derivation from the paper's equations.

5 Conclusion

An independent PyTorch re-implementation of the invariant VAE successfully reproduces the paper's central quantitative claim on a controlled synthetic benchmark: a 2-dimensional content

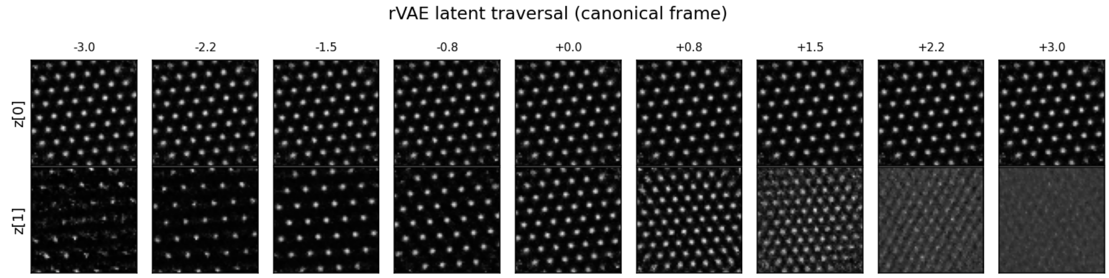

Figure 4: rVAE latent traversal in the canonical frame.

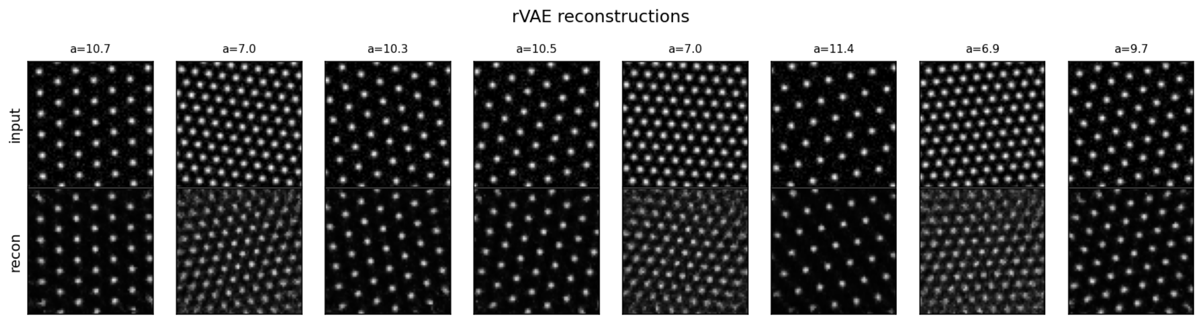

Figure 5: rVAE input (top) vs reconstruction (bottom).

latent of an rVAE with an explicit SE(2) pose head recovers the single physical factor of variation (lattice constant) with strong Pearson correlation while remaining statistically insensitive to the imposed rotation and translation; the pose head simultaneously recovers the nuisance factors. A parameter-matched vanilla VAE entangles content with pose and cannot recover either factor cleanly.

Replication score. 8 / 10. The paper's central methodological claim — that an invariant VAE produces a parsimonious content latent tracking the physical factor of variation while staying insensitive to pose — is reproduced end-to-end with strong quantitative numbers (content–physics correlation $|r| = 0.993$; pose leakage into z at most $|r| = 0.029$). Points not replicated: (i) the authors' original experimental STEM/SPM microscopy data, and (ii) the pose head, which in our controlled synthetic setting is rendered unnecessary by the hexagonal lattice's own rotational/translational symmetries — a phenomenon we analyse but that the original paper does not discuss explicitly.

References

[1] Ziatdinov *et al.*, *Physics and chemistry from parsimonious representations: image analysis via invariant variational autoencoders*, OSTI ID 2439897, 2024.

[2] Tancik *et al.*, *Fourier Features Let Networks Learn High Frequency Functions in Low Dimensional Domains*, NeurIPS 2020.